```
#include <iostream>
#include <string>
#include <ctime>
#include <cmath>

using namespace std;

class ParkingSlot {
public:
    int slotNumber;
    bool occupied;
    string vehicleNumber;
    string vehicleType;
    time_t entryTime;

    ParkingSlot(int number) {
        slotNumber = number;
        occupied = false;
        vehicleNumber = "";
        vehicleType = "";
        entryTime = 0;
    }

    void parkVehicle(string number, string type) {
        occupied = true;
        vehicleNumber = number;
        vehicleType = type;
        entryTime = time(nullptr);
    }
```

```
double removeVehicle() {
    time_t exitTime = time(nullptr);

    double hours =
        difftime(exitTime, entryTime) / (60 * 60);

    hours = max(1.0, ceil(hours));

    double rate;

    if (vehicleType == "Bike" ||
        vehicleType == "bike" ||
        vehicleType == "BIKE") {

        rate = 20;
    }
    else {
        rate = 40;
    }

    double fee = hours * rate;

    occupied = false;
    vehicleNumber = "";
    vehicleType = "";
    entryTime = 0;

    return fee;
```

```
    }
};

class SmartParkingSystem {
private:
    static const int TOTAL_SLOTS = 10;

    ParkingSlot* slots[TOTAL_SLOTS];

public:

    SmartParkingSystem() {
        for (int i = 0; i < TOTAL_SLOTS; i++) {
            slots[i] = new ParkingSlot(i + 1);
        }
    }

    ~SmartParkingSystem() {
        for (int i = 0; i < TOTAL_SLOTS; i++) {
            delete slots[i];
        }
    }

    void displaySlots() {

        cout << "\n========================================\n";
        cout << "          PARKING SLOT STATUS\n";
        cout << "========================================\n";
```

```
    for (int i = 0; i < TOTAL_SLOTS; i++) {

        if (slots[i]->occupied) {

            cout << "Slot " << slots[i]->slotNumber
                 << " -> OCCUPIED"
                 << " | Vehicle: "
                 << slots[i]->vehicleNumber
                 << " | Type: "
                 << slots[i]->vehicleType
                 << endl;
        }
        else {

            cout << "Slot "
                 << slots[i]->slotNumber
                 << " -> AVAILABLE"
                 << endl;
        }
    }
}

void parkVehicle() {

    int availableSlot = -1;

    for (int i = 0; i < TOTAL_SLOTS; i++) {

        if (!slots[i]->occupied) {
```

```
            availableSlot = i;
            break;
        }
    }

    if (availableSlot == -1) {
        cout << "\nSorry! Parking is FULL.\n";
        return;
    }

    string vehicleNumber;
    string vehicleType;

    cout << "\nEnter vehicle number: ";
    cin >> vehicleNumber;

    cout << "Enter vehicle type (Bike/Car): ";
    cin >> vehicleType;

    slots[availableSlot]->parkVehicle(
        vehicleNumber,
        vehicleType
    );

    cout << "\nVehicle parked successfully!\n";

    cout << "Assigned Parking Slot: "
         << slots[availableSlot]->slotNumber
         << endl;
```

```
}

void removeVehicle() {

    string vehicleNumber;

    cout << "\nEnter vehicle number: ";
    cin >> vehicleNumber;

    for (int i = 0; i < TOTAL_SLOTS; i++) {

        if (slots[i]->occupied &&
            slots[i]->vehicleNumber == vehicleNumber) {

            double fee = slots[i]->removeVehicle();

            cout << "\nVehicle removed successfully!\n";

            cout << "Parking Slot: "
                 << slots[i]->slotNumber
                 << endl;

            cout << "Parking Fee: Rs. "
                 << fee
                 << endl;

            return;
        }
    }
```

```
    cout << "\nVehicle not found!\n";
}

void searchVehicle() {

    string vehicleNumber;

    cout << "\nEnter vehicle number: ";
    cin >> vehicleNumber;

    for (int i = 0; i < TOTAL_SLOTS; i++) {

        if (slots[i]->occupied &&
            slots[i]->vehicleNumber == vehicleNumber) {

            cout << "\n========================================\n";
            cout << "            VEHICLE FOUND\n";
            cout << "========================================\n";

            cout << "Vehicle Number: "
                 << slots[i]->vehicleNumber
                 << endl;

            cout << "Vehicle Type: "
                 << slots[i]->vehicleType
                 << endl;

            cout << "Parking Slot: "
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```
                << slots[i]->slotNumber
                << endl;

            return;
        }
    }

    cout << "\nVehicle not found!\n";
}

void showAvailableSlots() {

    int count = 0;

    cout << "\nAvailable Parking Slots:\n";

    for (int i = 0; i < TOTAL_SLOTS; i++) {

        if (!slots[i]->occupied) {

            cout << slots[i]->slotNumber << " ";
            count++;
        }
    }

    cout << "\n\nTotal Available Slots: "
         << count
         << endl;
}
```

```
void menu() {

    int choice;

    do {

        cout << "\n\n";
        cout << "========================================\n";
        cout << "       SMART PARKING SLOT SYSTEM\n";
        cout << "========================================\n";

        cout << "1. Display Parking Slots\n";
        cout << "2. Park Vehicle\n";
        cout << "3. Remove Vehicle\n";
        cout << "4. Search Vehicle\n";
        cout << "5. Show Available Slots\n";
        cout << "6. Exit\n";

        cout << "========================================\n";

        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {

            case 1:
                displaySlots();
                break;
```

```
            case 2:
                parkVehicle();
                break;

            case 3:
                removeVehicle();
                break;

            case 4:
                searchVehicle();
                break;

            case 5:
                showAvailableSlots();
                break;

            case 6:
                cout << "\nThank you for using Smart Parking System!\n";
                break;

            default:
                cout << "\nInvalid choice! Please try again.\n";
        }

    } while (choice != 6);
  }
};
```

```
int main() {

    SmartParkingSystem system;

    system.menu();

    return 0;
}
```